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other attractor points in random matrix theory



In Random Matrix Theory (RMT) and its applications to dynamical systems, "attractor points" (or more broadly, attractors) represent stable states or patterns that the system converges toward over time. While traditional RMT focuses on spectral properties like eigenvalue distributions, the intersection with dynamical systems identifies several specific types of attractors: [R⁶ ResearchGate +2](#)

1. Arithmetic Attractor Points

In string theory and algebraic geometry, attractor points can have number-theoretic properties. For specific configurations (like Type IIB theory on certain Calabi-Yau manifolds), vector multiplets flow toward fixed points known as **attractors**. In cases like the T^6 or $K^3 \times T^2$ tori, these points correspond to **complex multiplication (CM)**, where the attractor points are considered "arithmetic" due to their unique algebraic properties. [X arXiv](#)

2. High-Dimensional Stationary Points

RMT is used to analyze the landscape of high-dimensional random fields (like those found in glass physics or neural networks). [R⁶ ResearchGate +1](#)

- **Stable Fixed Points:** These are the simplest attractors where the system state converges to a single constant value. [W Wikipedia +3](#)
- **Topology Trivialization:** A phenomenon where, as a control parameter (like noise or interaction strength) changes, a system with exponentially many stationary points suddenly collapses to just a few stable attractor points. [R⁶ ResearchGate](#)

3. Non-Fixed Point Attractors

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- **Torus Attractors:** Systems with multiple independent frequencies can produce a "doughnut-shaped" attractor. University of Babylon +1

4. Spectral "Attractors" (BBP Transition)

In high-dimensional statistics and machine learning, a key connection to RMT is the **BBP (Baik-Ben Arous-Péché) transition**. YouTube

- This transition describes how a "signal" (a specific eigenvalue) pops out from the "noise" (the bulk of the random matrix spectrum).
- In optimization dynamics, the system essentially "attracts" toward these outlier spectral directions, which act as summary statistics for the entire high-dimensional system. YouTube

5. Manifold Attractors in Neural Networks

In computational neuroscience, RMT helps explain how networks of neurons can restrict their activity to **low-dimensional manifolds**. YouTube

- **Continuous Attractors:** Rather than a single point, the system may be attracted to a continuous line or ring (e.g., a "line attractor"), which can represent continuous variables like an animal's position in space. YouTube +1



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give me open-sourced/freely availble books or summaries to
dive deeper into these



To dig deeper and more rigorously into these "attractor points" (universal limit laws) in Random Matrix Theory, you should look into the following open-access and freely available resources.

1. The "Gold Standard" Textbooks (Freely Available)

These provide the mathematical proof that these distributions are the "attractors" for large matrices.

- **"Topics in Random Matrix Theory" by Terence Tao (PDF available on his blog/UCLA site):**
 - **Why it's great:** Tao is a master communicator. This book rigorously covers the **Wigner Semicircle Law**, the **Marchenko-Pastur Law**, and **Universality** (why these distributions act as attractors regardless of the underlying entry distribution).
- **"Introduction to Random Matrices" by Greg Anderson, Alice Guionnet, and Ofer Zeitouni (PDF available on authors' websites):**
 - **Why it's great:** This is the rigorous "bible" for the subject. It focuses on the **Large Deviation Principle**, which mathematically defines how systems "settle" into these attractor states.

2. Deep Dives into Specific "Attractors"

- **For Tracy-Widom and Edge Scaling:**



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High-dimensional random matrices leads to a new "attractor" distribution.

3. Accessible Summaries & Lecture Notes

Attractor network - Wikipedia

An attractor network is a type of recurrent dynamical network, that evolves toward a stable pattern over time. Nodes in the attractor

- **MIT OpenCourseWare (18.338 Eigenvalues of Random Matrices):** Alan Edelman's course is legendary. The notes and Julia-based labs allow you to

see the eigenvalues "flowing" toward the Semicircle or Marchenko-Pastur

High-Dimensional Random Fields and Random Matrix Theory attractors.

Those works demonstrated that calculating the mean number of stationary points in such a setting can be mapped onto a random matrix

- **The IAS/Park City Mathematics Series:** Many of their volumes on RMT are available as pre-prints or open chapters. They bridge the gap between physics (chaos) and math (universality).

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4. Specialized Focus: Topology Trivialization

If you are interested in the **Algebraic/Geometric Attractors** mentioned in your previous search:

- Search arXiv for "**Complexity of Random Landscapes**" by **Gérard Ben Arous**. His papers explain how high-dimensional systems (like the loss surface of a neural network) have a "topology" that simplifies into a few attractor points as the system grows.

Quick Search Strategy

To find these immediately, use these exact strings in a search engine:

1. "Terence Tao" Random Matrix Theory PDF
2. "Speicher" Free Probability Lecture Notes PDF
3. "Edelman" MIT Random Matrix Course Notes
4. arXiv:1712.07903 (A great review paper on the universality of RMT).



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,
) approaches infinity.

arXiv

arXiv

+1

1. Statistical Attractors (Universality Classes)

In the same way the Gaussian distribution is an attractor for sums of random variables (Central Limit Theorem), RMT has its own "attractors" for eigenvalue distributions:

Wigner Semicircle Law: The global attractor for the density of states of Wigner matrices (real symmetric or Hermitian matrices with i.i.d. entries).

Marchenko-Pastur Law: The attractor for the singular value distribution of large rectangular matrices, commonly used in sample covariance analysis.

Tracy-Widom Distribution: An attractor for the fluctuations of the largest eigenvalue at the edge of the spectrum.

arXiv

arXiv

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2. Physical & Geometric Attractor Points

Beyond statistical distributions, "attractor points" appear when RMT intersects with dynamical systems and string theory:

Arithmetic Attractor Points: In string theory (Type IIB compactifications), vector multiplet moduli flow toward fixed points called attractors. At these specific points, the underlying Calabi-Yau manifold often has number-theoretic properties like complex multiplication (CM).

Neural Network Attractors: In computational neuroscience, RMT is used to analyze the stability of "attractor networks."

Large random connectivity matrices can create stable fixed points (static patterns) or limit cycles (periodic patterns) that model associative memory.

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exponents, meaning they are not chaotic despite their complex geometry.

Random Point/Set Attractors: In random dynamical systems, attractors are not fixed in space but are "random sets" that attract trajectories under stochastic influence.

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Would you like to explore the mathematical proofs behind the Tracy-Widom attractor or see how attractor networks are modeled using random connectivity?

. give me a summary as bullet heads of everything covered in this conversation ^

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